

```
CREATE DATABASE university_db;
```

```
USE university_db;
```

```
CREATE TABLE students (
```

```
    student_id INT PRIMARY KEY,
```

```
    name VARCHAR(100),
```

```
    email VARCHAR(100),
```

```
    gpa DECIMAL(3,2)
```

```
);
```

```
-- Populate 10 rows into students table
```

```
INSERT INTO students (student_id, name, email, gpa) VALUES
```

```
(1, 'Aung Aung', 'aung.aung@example.com', 3.45),
```

```
(2, 'Su Su', 'su.su@example.com', 3.80),
```

```
(3, 'Kyaw Kyaw', 'kyaw.kyaw@example.com', 2.95),
```

```
(4, 'Hla Hla', 'hla.hla@example.com', 3.20),
```

```
(5, 'Mya Mya', 'mya.mya@example.com', 3.90),
```

```
(6, 'Zaw Zaw', 'zaw.zaw@example.com', 2.75),
```

```
(7, 'Thida', 'thida@example.com', 3.60),
```

```
(8, 'Min Min', 'min.min@example.com', 3.10),
```

```
(9, 'Ei Ei', 'ei.ei@example.com', 3.50),
```

```
(10, 'Ko Ko', 'ko.ko@example.com', 2.85);
```

```
CREATE USER 'adminuser'@'localhost' IDENTIFIED BY 'AdminPass123';
```

```
CREATE USER 'readonlyuser'@'localhost' IDENTIFIED BY 'ReadOnly123';
```

```
CREATE USER 'dataentry'@'localhost' IDENTIFIED BY 'Entry123';
```

```
GRANT ALL PRIVILEGES ON university_db.* TO 'adminuser'@'localhost';

GRANT SELECT ON university_db.students TO 'readonlyuser'@'localhost';

GRANT INSERT, UPDATE ON university_db.students TO 'dataentry'@'localhost';
```

-- View Permissions

-- Check what privileges each user has:

34 • **SHOW GRANTS FOR 'adminuser'@'localhost';**

Result Grid Filter Rows: Export: Wrap Cell Content:	
Grants for adminuser@localhost	
▶	GRANT USAGE ON *.* TO `adminuse...
	GRANT ALL PRIVILEGES ON `universi...

35 • **SHOW GRANTS FOR 'readonlyuser'@'localhost';**

Result Grid Filter Rows: Export: Wrap Cell Content:	
Grants for readonlyuser@localhost	
▶	GRANT USAGE ON *.* TO `readonly...
	GRANT SELECT ON `university_db`

36 • **SHOW GRANTS FOR 'dataentry'@'localhost';**

Result Grid Filter Rows: Export: Wrap Cell Content:	
Grants for dataentry@localhost	
▶	GRANT USAGE ON *.* TO `dataentr...
	GRANT INSERT, UPDATE ON `univer...

-- Revoke a Privilege

```
REVOKE UPDATE ON university_db.students FROM 'dataentry'@'localhost';
```

-- Clean Up (Optional)

-- To remove all users after the test:

```
DROP USER 'adminuser'@'localhost';
```

DROP USER 'readonlyuser'@'localhost';

DROP USER 'dataentry'@'localhost';

DROP DATABASE university_db;

admin_user_testing x

Limit to 1000 rows

```
1 -- Testing for adminuser'@'localhost' IDENTIFIED BY 'AdminPass123';
2 • USE university_db;
3 • SELECT * FROM students;
4
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	student_id	name	email	gpa
▶	1	Aung Aung	aung.aung@example.com	3.45
	2	Su Su	su.su@example.com	3.80
	3	Kyaw Kyaw	kyaw.kyaw@example.com	2.95
	4	Hla Hla	hla.hla@example.com	3.20
	5	Mya Mya	mya.mya@example.com	3.90
	6	Zaw Zaw	zaw.zaw@example.com	2.75
	7	Thida	thida@example.com	3.60
	8	Min Min	min.min@example.com	3.10
	9	Ei Ei	ei.ei@example.com	3.50
	10	Ko Ko	ko.ko@example.com	2.85
	13	Soe Soe	soesoe@example.com	3.55
*	NULL	NULL	NULL	NULL

```
5 • INSERT INTO students (student_id, name, email, gpa) VALUES
6 (11, 'Aung Aung Moe', 'aung.aungmoe@example.com', 3.55);
7
8 • UPDATE students
9 SET email='aungmoe@example.com'
10 WHERE student_id=11;
11
12 • DELETE FROM students
13 WHERE student_id=11;
14
```

Output

Action Output

#	Time	Action
✓ 1	21:10:42	USE university_db
✓ 2	21:10:44	SELECT * FROM students LIMIT 0, 1000
✓ 3	21:10:56	INSERT INTO students (student_id, name, email, gpa) VALUES (11, 'Aung Aung Moe', 'aung.aungmoe@exa...
✓ 4	21:10:57	UPDATE students SET email='aungmoe@example.com' WHERE student_id=11
✓ 5	21:10:58	DELETE FROM students WHERE student_id=11

entry_user_testing x

Limit to 1000 r

```
1  -- Testing for 'dataentry'@'localhost' IDENTIFIED BY 'Entry123';
2 •  USE university_db;
3  -- error
4 •  SELECT * FROM students;
5
6 •  INSERT INTO students (student_id, name, email, gpa) VALUES
7  (13, 'Soe Soe', 'soesoe@example.com', 3.55);
8
9 •  UPDATE students
10 SET email='aungmoe@example.com'
11 WHERE student_id=13;
12 -- error
13 •  DELETE FROM students
14 WHERE student_id=13;
15
```

Output

Action Output

	#	Time	Action
✓	1	21:12:58	USE university_db
✗	2	21:12:59	SELECT * FROM students LIMIT 0, 1000
✓	3	21:13:03	INSERT INTO students (student_id, name, email, gpa) VALUES (13, 'Soe Soe', 'soesoe@example.com', 3.55)
✗	4	21:13:06	UPDATE students SET email='aungmoe@example.com' WHERE student_id=13
✗	5	21:13:12	DELETE FROM students WHERE student_id=13

```
1  -- Testing for 'readonlyuser'@'localhost' IDENTIFIED BY 'ReadOnly123';
2  •  USE university_db;
3  •  SELECT * FROM students;
4
```

Result Grid | Filter Rows: | Edit: | Export: Import: | Wrap Cell Content:

	student_id	name	email	gpa
▶	1	Aung Aung	aung.aung@example.com	3.45
	2	Su Su	su.su@example.com	3.80
	3	Kyaw Kyaw	kyaw.kyaw@example.com	2.95
	4	Hla Hla	hla.hla@example.com	3.20
	5	Mya Mya	mya.mya@example.com	3.90
	6	Zaw Zaw	zaw.zaw@example.com	2.75
	7	Thida	thida@example.com	3.60
	8	Min Min	min.min@example.com	3.10
	9	Ei Ei	ei.ei@example.com	3.50
	10	Ko Ko	ko.ko@example.com	2.85
	13	Soe Soe	soesoe@example.com	3.55
*	NULL	NULL	NULL	NULL

```

5  -- error
6 • INSERT INTO students (student_id, name, email, gpa) VALUES
7  (11, 'Aung Aung Moe', 'aung.aungmoe@example.com', 3.55);
8
9  -- error
10 • UPDATE students
11  SET email='aungmoe@example.com'
12  WHERE student_id=10;
13  -- error
14 • DELETE FROM students
15  WHERE student_id=10;
16  -- error
17 • ALTER TABLE students ADD COLUMN address VARCHAR(150);
18  -- can view
19 • DESC students;

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	Field	Type	Null	Key	Default	Extra
▶	student_id	int	NO	PRI	NULL	
	name	varchar(100)	YES		NULL	
	email	varchar(100)	YES		NULL	
	gpa	decimal(3,2)	YES		NULL	

Result 2

Output

Action Output

	#	Time	Action
✓	1	21:15:17	USE university_db
✓	2	21:15:20	SELECT * FROM students LIMIT 0, 1000
✗	3	21:15:31	INSERT INTO students (student_id, name, email, gpa) VALUES (11, 'Aung Aung Moe', 'aung.aungmoe@exa...
✗	4	21:15:33	UPDATE students SET email='aungmoe@example.com' WHERE student_id=10
✗	5	21:15:34	DELETE FROM students WHERE student_id=10
✗	6	21:15:36	ALTER TABLE students ADD COLUMN address VARCHAR(150)
✓	7	21:15:41	DESC students